

NGUYEN HOANG SANG

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Professional Summary

Final-year Software Engineering student specializing in Backend Development (Spring Boot, Node.js, Microservices). Experienced in engineering robust server-side architectures, high-performance RESTful APIs, and scalable event-driven layers. Seeking a Backend Developer Intern role to contribute to production-grade software products and optimize system integration workflows.

Technical Skills

Core Languages: Java, JavaScript, TypeScript, SQL

Backend Frameworks: Spring Boot, Spring Security, Spring Data JPA, Node.js, Express.js

Databases: Redis (Caching & Session), MongoDB, MariaDB, SQL Server

Messaging & Real-time: RabbitMQ (Message Broker), Socket.IO, WebSockets, LiveKit SDK

Cloud & DevOps: Docker, GitHub Actions (CI/CD), AWS (EC2, S3, ECR, SSM), JWT, OAuth2

Education

Industrial University of Ho Chi Minh City (IUH)
Software Engineering — Faculty of Information Technology

2022–Present
GPA: 3.12 / 4.0

Awards

Academic Encouragement Scholarship
Industrial University of Ho Chi Minh City (IUH) — Excellent Academic Performance

2023–2024

Projects

Riff — Realtime OTT Chat Platform

05/2026–06/2026

Node.js, Express, Socket.IO, LiveKit SDK, MongoDB, Redis, RabbitMQ, Spring Boot, Docker, AWS

Links: [Backend](#) — [Live Demo](#)

- Co-engineered a production-grade OTT messaging application utilizing a 7-service microservices architecture.
- Engineered the core Chat microservice with Node.js and Socket.IO to enable low-latency event processing.
- Designed optimized MongoDB document collections and compound indexes to handle high-throughput chat histories.
- Integrated an abstract object storage layer using AWS S3 to securely serve and manage multi-format attachments.
- Programmed real-time server events for instant message delivery handshakes, message reactions, and user typing states.
- Utilized Redis and Socket.IO Redis Adapter to maintain cross-server WebSocket session synchronization.
- Orchestrated asynchronous messaging pipelines using RabbitMQ to decouple computationally intensive notification tasks.
- Automated path-dependent container deployment to AWS EC2 nodes via GitHub Actions and AWS SSM.

Shop Perfume — Full-stack E-commerce Infrastructure

12/2025–05/2026

Java 17, Spring Boot 3, Spring Security, Spring Data JPA, MariaDB, Redis, SePay Webhook

Links: [GitHub Source](#) — [Live Demo](#)

- Developed modular, type-safe RESTful APIs utilizing Java 17 and Spring Boot 3 with Spring Data JPA filters.
- Enforced strict stateless authorization boundaries within Spring Security via native JWT and Google OAuth2 integration.
- Achieved 100% payment verification automation by implementing an API webhook controller with SePay gateway.
- Formulated backend core logic for inventory catalogs, persistent shopping carts, and relational checkout tables.
- Deployed Redis memory caching layers to optimize expensive relational queries and store temporary verification codes.
- Integrated Cloudinary server SDK endpoints to handle remote multimedia storage and dynamic asset processing.